PADHAKU
Contemporary India · Class 10 Social Science



CHAPTER 3

# Water Resources

NCERT Exercise Solutions


Activity · Water Scarcity and the Need for Water Conservation and Management — 1 Q

Activity · Multi-Purpose River Projects and Integrated Water Resources Management — 3 Q

Activity · Rainwater Harvesting — 4 Q

Exercises — 10 Q


✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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## Activity · Water Scarcity and the Need for Water Conservation and Management

**1** **From your everyday experiences, write a short proposal on how you can conserve water.**

💡 SAMPLE ANSWER

### Proposal: Saving Water at Home and School

- Turn off the tap while brushing teeth, soaping hands or washing utensils.
- Use a bucket and mug for bathing instead of a shower or running hose.
- Repair leaking taps, pipes and flush tanks at once.
- Reuse water: water used for washing vegetables or RO waste water can water plants or clean floors.
- Wash vehicles with a bucket and cloth, not a running pipe.
- Water plants in the early morning or evening so less water evaporates.
- Set up rooftop rainwater harvesting at home or school to collect rain and recharge groundwater.
- Make friends and neighbours aware through posters and a 'Save Water' club in school.

**Conclusion:** Small daily habits by every family can save thousands of litres of water every year.

## Activity · Multi-Purpose River Projects and Integrated Water Resources Management

**1** **Find out more about any one traditional method of building dams and irrigation works.**

SAMPLE ANSWER

**Kallanai (Grand Anicut), Tamil Nadu**

- Built across the river Kaveri by the Chola king Karikala, nearly 2,000 years ago.
- Made of large uncut stones placed across the river bed and joined with clay; no cement was used.
- It diverts river water into canals that irrigate fields of the Kaveri delta.
- It is still in use today and is one of the oldest water-diversion structures in the world.

Other traditional methods: guls/kuls of the Western Himalayas, khadins and johads of Rajasthan, inundation channels of Bengal.

**2** **Collect information about floods occurred in different parts of the country due to heavy rainfall in recent times.**

SAMPLE ANSWER

Some major floods caused by heavy rain:

- **Uttarakhand (June 2013):** cloudburst and very heavy rain; huge loss of life around Kedarnath.

- **Chennai, Tamil Nadu (December 2015):** record rainfall flooded the city.
- **Kerala (August 2018):** worst floods in about a century; most districts affected.
- **Himachal Pradesh (July–August 2023):** very heavy monsoon rain caused floods and landslides; roads and bridges washed away.
- **Wayanad, Kerala (July 2024):** extremely heavy rain caused deadly landslides.
- **Assam and Bihar:** floods occur almost every year in the Brahmaputra and Ganga plains.

Effects: loss of life and property, crops destroyed, soil erosion, spread of water-borne diseases. (Add floods from your own state from newspapers.)

### 3 Make a list of inter-state water disputes.

💡 SAMPLE ANSWER

| River / project | States involved |
|---|---|
| Cauvery (Kaveri) | Karnataka, Tamil Nadu, Kerala, Puducherry |
| Krishna | Maharashtra, Karnataka, Andhra Pradesh, Telangana |
| Godavari | Maharashtra, Andhra Pradesh, Karnataka, Madhya Pradesh, Odisha |
| Narmada | Gujarat, Madhya Pradesh, Maharashtra, Rajasthan |
| Ravi–Beas | Punjab, Haryana, Rajasthan |

| | |
|---|---|
| Sutlej–Yamuna Link canal | Punjab, Haryana |
| Mahadayi (Mandovi) | Goa, Karnataka, Maharashtra |
| Vansadhara | Odisha, Andhra Pradesh |
| Mahanadi | Odisha, Chhattisgarh |
| Mullaperiyar dam | Kerala, Tamil Nadu |

## 🧪 Activity · Rainwater Harvesting

**1** **Collect information about flood prone areas of the country.**

💡 SAMPLE ANSWER

- **Ganga plains:** Uttar Pradesh, Bihar, West Bengal (Kosi river is called the 'sorrow of Bihar').
- **Brahmaputra valley:** Assam and other north-eastern states – floods almost every monsoon.
- **East coast deltas:** Odisha (Mahanadi), Andhra Pradesh (Godavari, Krishna), Tamil Nadu – floods from cyclones and heavy rain.
- **Kerala and the Konkan coast:** very heavy monsoon rainfall.
- **Damodar valley (Jharkhand–West Bengal):** the Damodar was called the 'river of sorrow'.
- **Gujarat and Punjab–Haryana plains:** floods in years of very heavy rain.

About 40 million hectares of India's land is considered flood prone (Rashtriya Barh Ayog).

**2** **Find out other rainwater harvesting systems existing in and around your locality.**

💡 **SAMPLE ANSWER**

Model answer (look around your own area):

- **Rooftop rainwater harvesting:** pipes take rain from roofs to a tank or recharge pit (common in schools and government buildings).
- **Recharge pits / trenches:** pits filled with pebbles and sand let rainwater soak into the ground.
- **Recharge of old wells and hand pumps:** filtered rainwater is sent into an abandoned dugwell or around a hand pump.
- **Village ponds and tanks:** collect run-off from fields and streets.
- **Check dams** on small streams slow the water so it seeps into the ground.

Traditional systems in some areas: johads, khadins, tankas (Rajasthan), kuls (Himachal), bamboo drip irrigation (Meghalaya).

**3** **Collect information on how industries are polluting our water resources.**

💡 **SAMPLE ANSWER**

- Many factories release **untreated effluents** (waste water with chemicals) into rivers and lakes.
- **Tanneries** (e.g. at Kanpur on the Ganga) release chromium and other chemicals.
- **Textile dyeing and paper mills** add colours, bleach and chemicals to water.
- **Chemical, fertiliser and pesticide units** add toxic substances and heavy metals like mercury and lead.
- **Thermal power plants** release hot water (thermal pollution) and fly ash.
- **Mining** washes waste and metals into streams; oil refineries can cause oil spills.
- Industrial waste also seeps into the ground and pollutes **groundwater**.

Effects: fish die, water becomes unsafe to drink, and diseases spread. Solution: effluent treatment plants and strict laws.

**4** **Enact with your classmates a scene of water dispute in your locality.**

💡 SAMPLE ANSWER

**Sample scene: 'One Tap, Many Buckets'**

- **Characters:** two neighbours, a shopkeeper, a ward member, a student.
- **Scene 1:** Water comes in the colony tap for only one hour. Neighbours quarrel over who fills first; one uses a motor pump to pull more water.

- **Scene 2:** The ward member is called. Each side complains that the other wastes water (washing cars, overflowing tanks).
- **Scene 3:** The student suggests a fair turn system, a ban on direct motor pumps, fixing leaks and rooftop rainwater harvesting.
- **Ending:** Everyone agrees to share water fairly and save it. Message: unequal access and waste cause scarcity; cooperation solves it.

## 📝 Exercises

**1 (i)** **Classify each situation as 'suffering from water scarcity' or 'not suffering from water scarcity':**

**(a) Region with high annual rainfall.**

**(b) Region having high annual rainfall and large population.**

**(c) Region having high annual rainfall but water is highly polluted.**

**(d) Region having low rainfall and low population.**

✔ ANSWER

(a) **Not suffering** from water scarcity – enough rain to meet needs.

(b) **Suffering** from water scarcity – a large population creates very high demand and over-uses water.

(c) **Suffering** from water scarcity – polluted water is not fit for use (scarcity due to bad quality).

(d) **Not suffering** from water scarcity – with few people, the demand is low, so the little rain can meet it.

**1 (ii)** **Which one of the following statements is not an argument in favour of multi-purpose river projects?**

**(a) Multi-purpose projects bring water to those areas which suffer from water scarcity.**

**(b) Multi-purpose projects by regulating water flow helps to control floods.**

**(c) Multi-purpose projects lead to large scale displacements and loss of livelihood.**

**(d) Multi-purpose projects generate electricity for our industries and our homes.**

✔ ANSWER

**(c)** Multi-purpose projects lead to large scale displacements and loss of livelihood.

This is a disadvantage, so it is an argument against such projects.

**1 (iii) (a)** **Multiplying urban centres with large and dense populations and urban lifestyles have helped in proper utilisation of water resources.**

✔ ANSWER

**Mistake:** 'helped in proper utilisation'.

**Correct:** Multiplying urban centres with large and dense populations and urban lifestyles have added to water and energy requirements and led to over-exploitation of water resources.

**1 (iii) (b)** **Regulating and damming of rivers does not affect the river's natural flow and its sediment flow.**

✔ ANSWER

**Mistake:** 'does not affect'.

**Correct:** Regulating and damming of rivers affects the river's natural flow, causing poor sediment flow and excessive sedimentation at the bottom of the reservoir.

**1 (iii) (c)** **Today in Rajasthan, the practice of rooftop rainwater harvesting has gained popularity despite high water availability due to the Indira Gandhi Canal.**

✔ ANSWER

**Mistake:** 'has gained popularity despite'.

**Correct:** Today in western Rajasthan, the practice of rooftop rainwater harvesting is on the decline as plenty of water is available due to the perennial Indira Gandhi Canal.

**2 (i)** **Explain how water becomes a renewable resource.**

✔ ANSWER

- Freshwater comes mainly from surface run-off and groundwater.
- These are continually renewed and recharged through the **hydrological cycle** (evaporation, condensation, precipitation).
- All water moves within this cycle, so water used is replaced again. Hence water is a renewable resource.

**2 (ii)** **What is water scarcity and what are its main causes?**

✔ ANSWER

**Water scarcity** is the shortage of water (in quantity or of usable quality) to meet the needs of people.

**Causes:**

- Over-exploitation and excessive use of water.
- Unequal access to water among social groups.

- Large and growing population; more irrigation for food, industrialisation and urbanisation.
- Pollution of water, making it unfit for use.

**2 (iii)** **Compare the advantages and disadvantages of multi-purpose river projects.**

✔ ANSWER

| Advantages | Disadvantages |
|---|---|
| Irrigation for agriculture | Affect natural flow; poor sediment flow and sedimentation in reservoir |
| Hydroelectricity for industries and homes | Fragment rivers; harm aquatic life and fish migration |
| Water supply for domestic and industrial use | Submerge vegetation, soil and land; displace local people |
| Flood control, inland navigation, fish breeding, recreation | Can trigger floods, earthquakes, water-borne diseases; salinisation of soil |

**3 (i)** **Discuss how rainwater harvesting in semi-arid regions of Rajasthan is carried out.**

✔ ANSWER

In the semi-arid and arid regions of Rajasthan, people developed rooftop rainwater harvesting to store drinking water.

1. In places like **Bikaner, Phalodi and Barmer**, almost all houses traditionally had underground tanks called **tankas**.

2. Tankas were built inside the main house or courtyard and could be as large as a big room (one in Phalodi was 6.1 m deep, 4.27 m long and 2.44 m wide).

3. They were connected to the **sloping roofs** through a pipe; rain falling on the roof flowed down the pipe into the tanka.

4. The **first spell of rain was not collected**, as it cleaned the roofs and pipes; water from later showers was stored.

5. Stored rainwater, called **palar pani**, lasted till the next rainfall and was a reliable source of drinking water in summer; it is considered the purest form of natural water.

6. Many houses built **underground rooms** next to the tanka to keep cool in summer. Fields were also turned into rain-fed storage structures like **khadins** (Jaisalmer) and **johads**.

Today the practice is declining in western Rajasthan because of the Indira Gandhi Canal, though some houses keep tankas as they prefer the taste of rainwater.

**3 (ii)** **Describe how modern adaptations of traditional rainwater harvesting methods**

**are being carried out to conserve and store water.**

✔ ANSWER

Traditional rainwater harvesting methods are being adapted in many rural and urban areas of India today.

1. **Rooftop rainwater harvesting:** rain from rooftops is collected with PVC pipes, filtered through sand and bricks and stored in a sump for immediate use.

2. **Groundwater recharge:** extra water is sent to wells, abandoned dugwells or around hand pumps to recharge underground water.

3. **Gendathur village (Mysuru, Karnataka):** about 200 households installed rooftop systems; each house can collect about 50,000 litres a year, and the village is known as 'rich in rainwater'.

4. **Shillong (Meghalaya):** nearly every household has a rooftop system that meets 15–25% of its water needs, though it is near Cherrapunjee and Mawsynram.

5. **Tamil Nadu** was the first state to make rooftop rainwater harvesting compulsory for all houses, with legal punishment for defaulters.

6. **Bamboo drip irrigation (Meghalaya):** a 200-year-old system where bamboo pipes carry spring water to plants, delivering 20–80 drops a minute near the roots.

> Thus, old wisdom combined with modern materials helps conserve water and reduce scarcity.

🎉 **Chapter 3 done! Revise the answers once more tomorrow.**